Ericsson PE Node WHI-PE-14 Optical Transport Failure SOP (ALARM_7810)

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_7810

Description

This SOP outlines the procedures for diagnosing and resolving a major optical transport failure on Ericsson PE node WHI-PE-14 at the Whitefield East POP, as indicated by alarm code ALARM_7810. The PE node is equipped with a high-speed optical transport network, utilizing SFP/XFP transceivers to facilitate data transfer over the fiber optic cables. The Optical Transport Failure (OTF) domain encompasses a range of potential issues, including SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. These issues can arise due to various factors, such as hardware malfunctions, environmental stressors, or human error. The Ericsson PE node's role in the network is critical, as it acts as a gateway for data transmission between different network segments. The OTF failure on WHI-PE-14 can impact the overall network performance, resulting in service outages, packet loss, and increased latency. Therefore, prompt diagnosis and resolution of this issue are essential to ensure network stability and prevent potential cascading failures.

Pre-requisite Checks

- Verify that the Whitefield East POP has been secured and is accessible for remote hands to perform necessary checks and interventions.

- Confirm that the redundant link status is stable and not contributing to the OTF failure.

- Ensure that the necessary diagnostic tools and equipment are available at the POP.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show optical transport stats, Expected Output: Optical transport statistics, including SFP/XFP status, fiber utilization, and Rx/Tx power levels., Description: This command provides an overview of the optical transport network's performance, allowing for the identification of potential issues and areas for further investigation.

Command: show interface optics, Expected Output: Detailed interface optics information, including SFP/XFP diagnostics, fiber status, and alignment metrics., Description: This command offers in-depth
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information about the interface optics, enabling the detection of
SFP/XFP degradation, fiber cuts, or other issues affecting the optical
transport network.

Command: show optical transport alarms, Expected Output: List of
active optical transport alarms, including alarm code, description,
and timestamp., Description: This command displays a comprehensive
list of active optical transport alarms, allowing for the
identification of the specific alarm code (ALARM_7810 in this case)
and its corresponding description and timestamp.
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Resolution Steps

Step 1: Step: Verify SFP/XFP status, Command: show optical transport stats | include SFP, Description: Check the SFP/XFP status to determine if it is functioning within normal parameters. If the SFP/XFP is degraded, proceed with replacing it.

Step 2: Step: Replace SFP/XFP, Command: replace optical transport sfp/xfp , Description: Replace the degraded SFP/XFP with a new one, ensuring that the replacement is done in accordance with the manufacturer's instructions and industry best practices.

Step 3: Step: Verify fiber status, Command: show interface optics | include fiber, Description: Verify the fiber status to ensure that it is intact and functioning properly. If the fiber is cut, proceed with repairing or replacing it.

Step 4: Step: Repair/replace fiber, Command: repair/replace fiber optic cable , Description: Repair or replace the damaged fiber optic cable, ensuring that the replacement is done in accordance with the manufacturer's instructions and industry best practices.

Post-Check Verification

```
Command: show optical transport stats, Expected Output: Optical
transport statistics indicating normal operation., Description: Verify
that the optical transport network is operating within normal
parameters, with stable SFP/XFP status, fiber utilization, and Rx/Tx
power levels.

Command: show interface optics, Expected Output: Detailed interface
optics information indicating normal operation., Description: Verify
that the interface optics are functioning properly, with no signs of
SFP/XFP degradation, fiber cuts, or other issues.
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Roll-back Procedures

- Step: Restore original SFP/XFP, Command: restore optical transport sfp/xfp , Description: Restore the original SFP/XFP to its previous state, in case the replacement caused unintended consequences.

- Step: Verify original fiber status, Command: show interface optics | include fiber, Description: Verify that the original fiber status is intact and functioning properly, in case the replacement caused unintended consequences.